- C++

An AWS SDK is a convenient way to make calls to the AWS API from your favorite programming language. The SDK takes care of things like authentication, retry on error, HTTPS communication, and XML or JSON (de)serialization. You're free to choose the SDK for your favorite language, but in this book, most examples are written in JavaScript and run in the Node.js runtime environment.

Installing and getting started with Node.js

Node.js is a platform for executing JavaScript in an event-driven environment so you can easily build network applications. To install Node.js, visit https://nodejs.org and download the package that fits your OS. All examples in this book are tested with Node.js 14.

After Node.js is installed, you can verify that everything works by typing node --version into your terminal. Your terminal should respond with something similar to `v14.*`. Now you're ready to run JavaScript examples, like the Node Control Center for AWS. Your Node.js installation comes with a important tool called npm, which is the package manager for Node.js. Verify the installation by running `npm --version` in your terminal.

To run a JavaScript script in Node.js, enter `node script.js` in your terminal, where `script.js` is the name of the script file. We use Node.js in this book because it's easy to install, it requires no IDE, and the syntax is familiar to most programmers.

Don't be confused by the terms *JavaScript* and *Node.js*. If you want to be precise, JavaScript is the language and Node.js is the runtime environment. But don't expect anybody to make that distinction. Node.js is also called node.

Do you want to get started with Node.js? We recommend *Node.js in Action* (second edition) by Alex Young et al. (Manning, 2017), or the video course *Node.js in Motion* by PJ Evans (Manning, 2018).

To understand how the AWS SDK for Node.js (JavaScript) works, let's create a Node.js (JavaScript) application that controls EC2 instances via the AWS SDK. The name of the application is *Node Control Center for AWS* or *nodecc* for short.